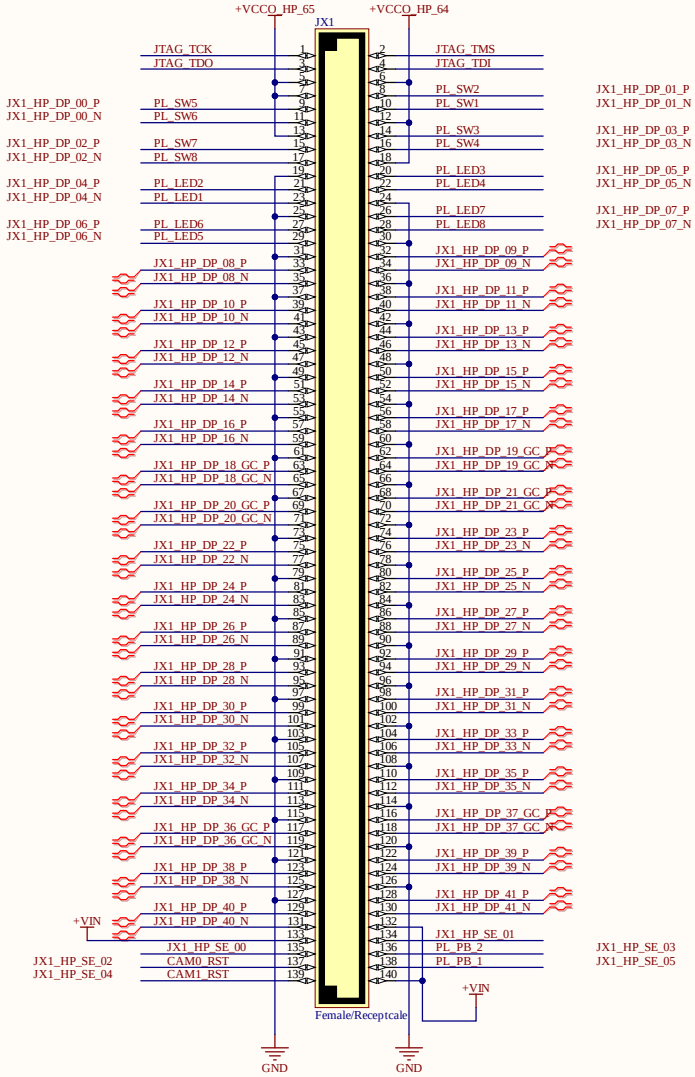
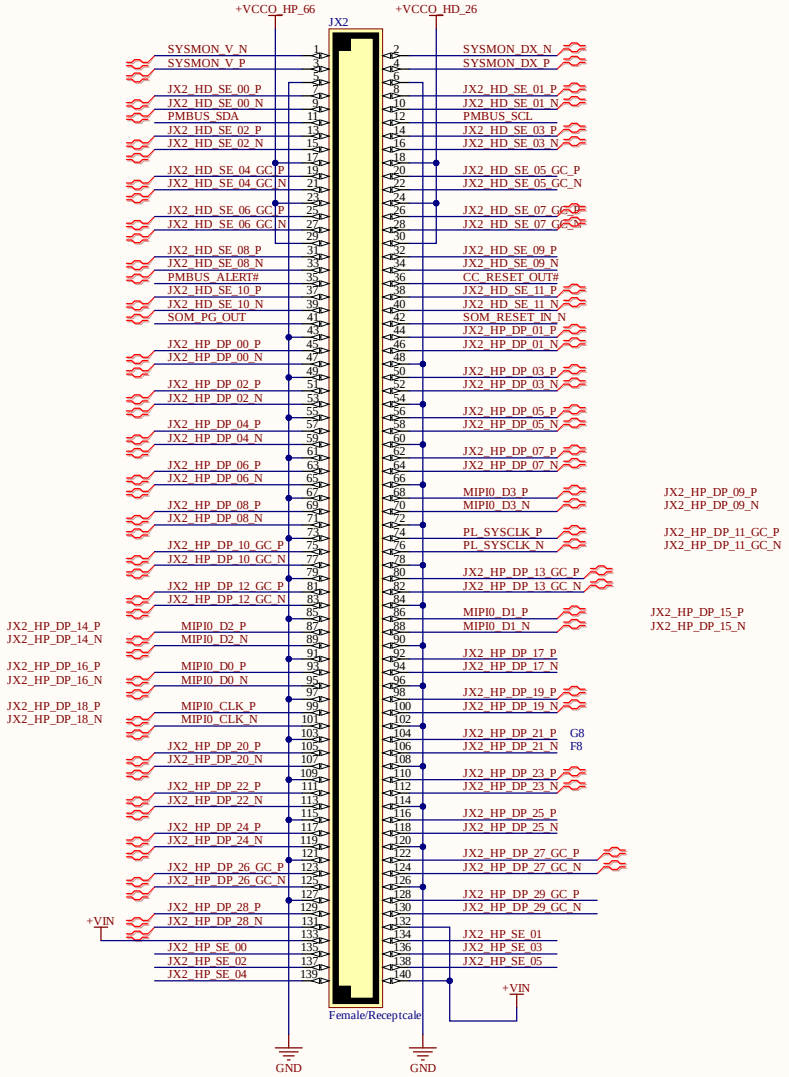
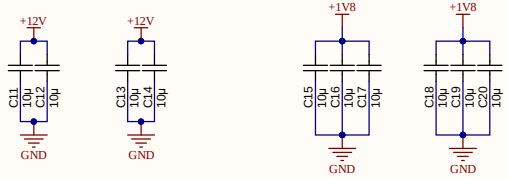
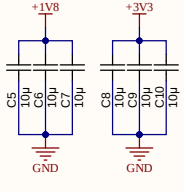
+VCCO_PSIO, 1V8, JTAG is +VCCO_PSIO

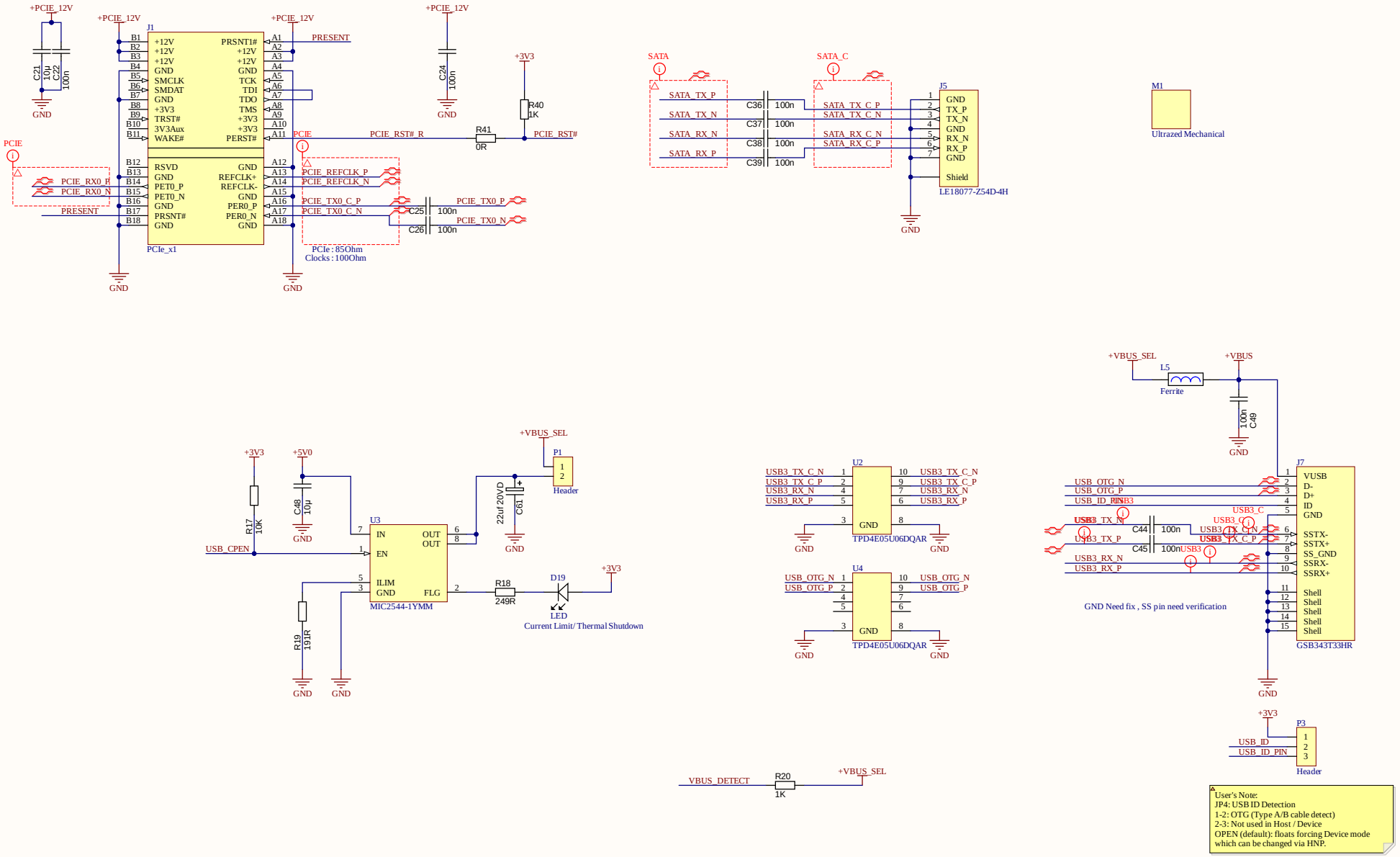


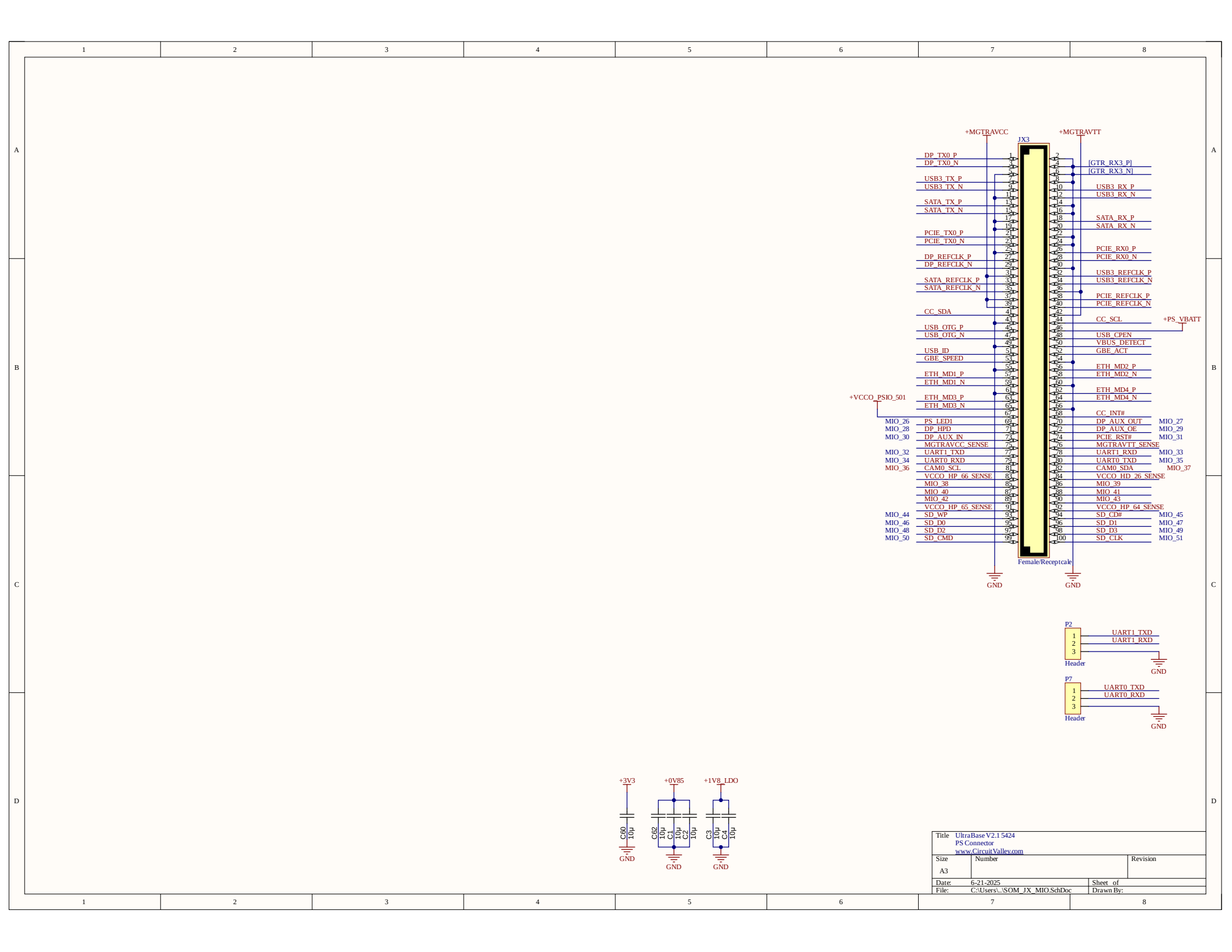



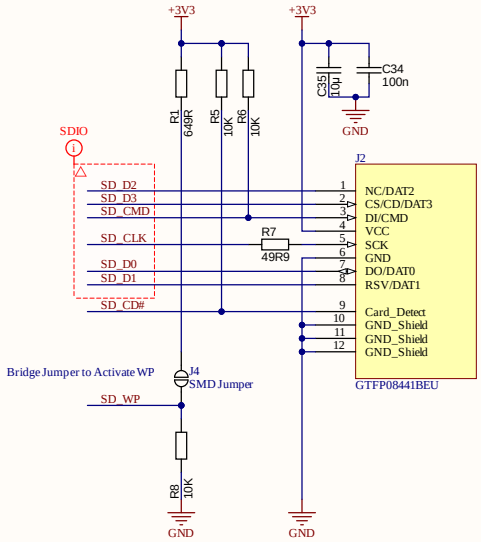



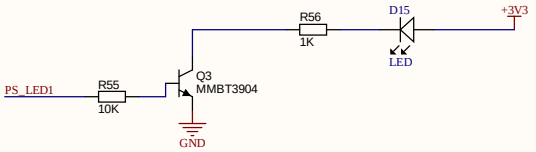

| | | |
|---|---|---|
| Title | UltraBase V2.1 5424<br>PL Connectors<br>www.CircuitValley.com | |
| Size<br>A3 | Number | Revision |
| Date: | 6-21-2025 | Sheet of |
| File: | C:\Users\..\PL_Conenctor.schDoc | Drawn By: |


+PCIE_12V
C21 10µ
C22 100n
GND
J1
+12V
SMCLK
SMDAT
+3V3
TRST#
3V3Aux
WAKE#
PRSNT1#
PRESENT
TCK
TDI
TDO
TMS
PERST#
RSVD
REFCLK+
REFCLK-
PER0_P
PER0_N
PCIE_RST# R
R41 0R
PCIE_RST#
R40 1K
+3V3
C24 100n
PCIE
PCIE_RX0_P
PCIE_RX0_N
PRESENT
PCIE_REFCLK_P
PCIE_REFCLK_N
PCIE_TX0_C_P
PCIE_TX0_C_N
C25 100n
C26 100n
PCIE_TX0_P
PCIE_TX0_N
PCIe_x1
PCIe : 85Ohm
Clocks : 100Ohm
SATA
SATA_TX_P
SATA_TX_N
SATA_RX_N
SATA_RX_P
C36 100n
C37 100n
C38 100n
C39 100n
SATA_C
SATA_TX_C_P
SATA_TX_C_N
SATA_RX_C_N
SATA_RX_C_P
J5
GND
TX_P
TX_N
RX_N
RX_P
Shield
LE18077-Z54D-4H
M1
Ultrazed Mechanical
+VBUS_SEL
L5
Ferrite
+VBUS
C49 100n
P1
Header
+3V3
+5V0
R17 10K
C48 10µ
22uf 20VD
C61
U3
IN
OUT
EN
ILIM
FLG
USB_CPEN
R18 249R
D19
LED
R19 191R
MIC2544-1YMM
Current Limit/ Thermal Shutdown
U2
USB3_TX_C_N
USB3_TX_C_P
USB3_RX_N
USB3_RX_P
TPD4E05U06DQAR
U4
USB_OTG_N
USB_OTG_P
J7
VUSB
D-
D+
ID
SSTX-
SSTX+
SS_GND
SSRX-
SSRX+
Shell
C44 100n
C45 100n
GSB343T33HR
GND Need fix , SS pin need verification
P3
USB_ID
USB_ID_PIN
VBUS_DETECT
R20 1K
User's Note:
JP4: USB ID Detection
1-2: OTG (Type A/B cable detect)
2-3: Not used in Host / Device
OPEN (default): floats forcing Device mode
which can be changed via HNP.
Title UltraBase V2.1 5424
PCIe, SATA USB OTG
www.CircuitValley.com
Size A3
Number
Revision
Date: 6-21-2025
File: C:\Users\..\PCIe.SchDoc
Sheet of
Drawn By:

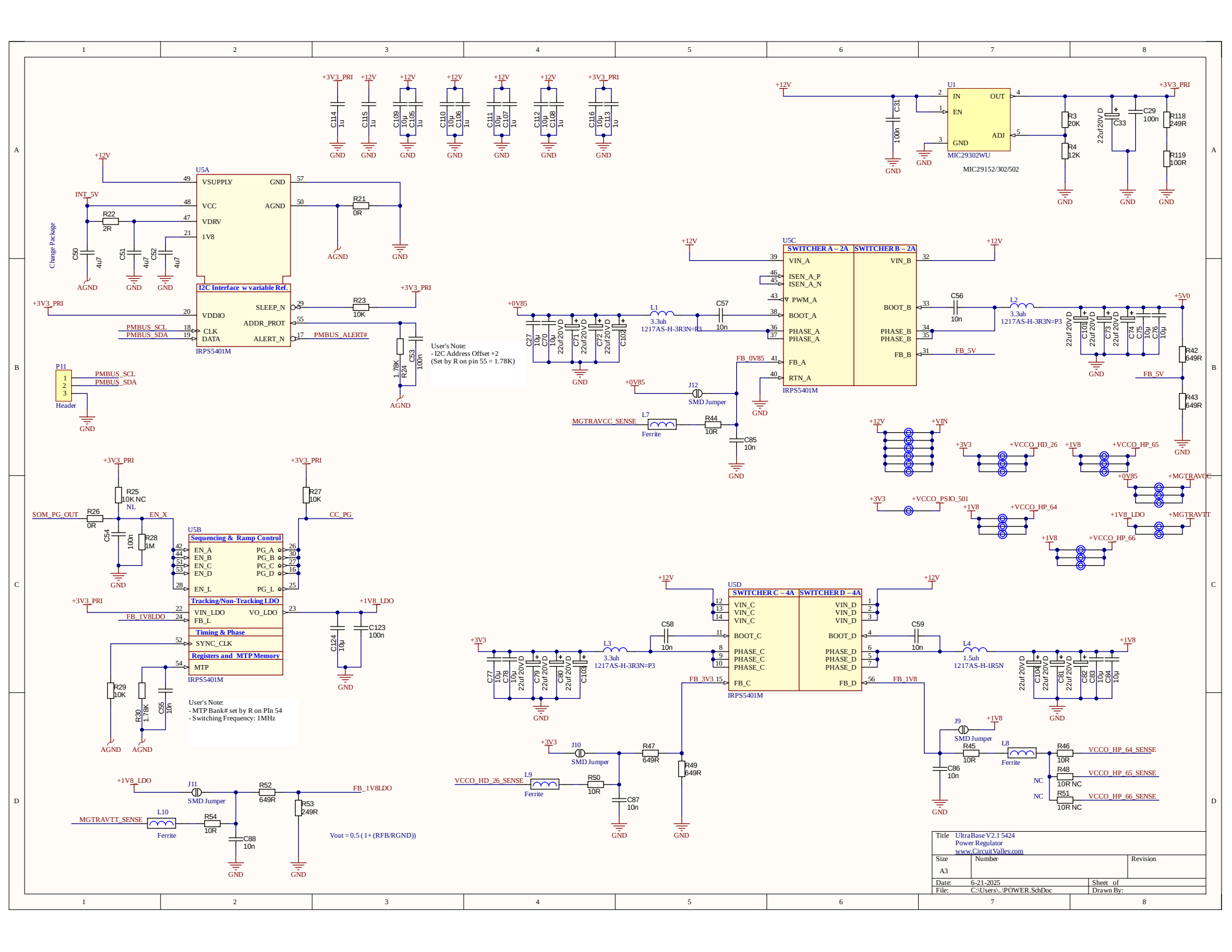


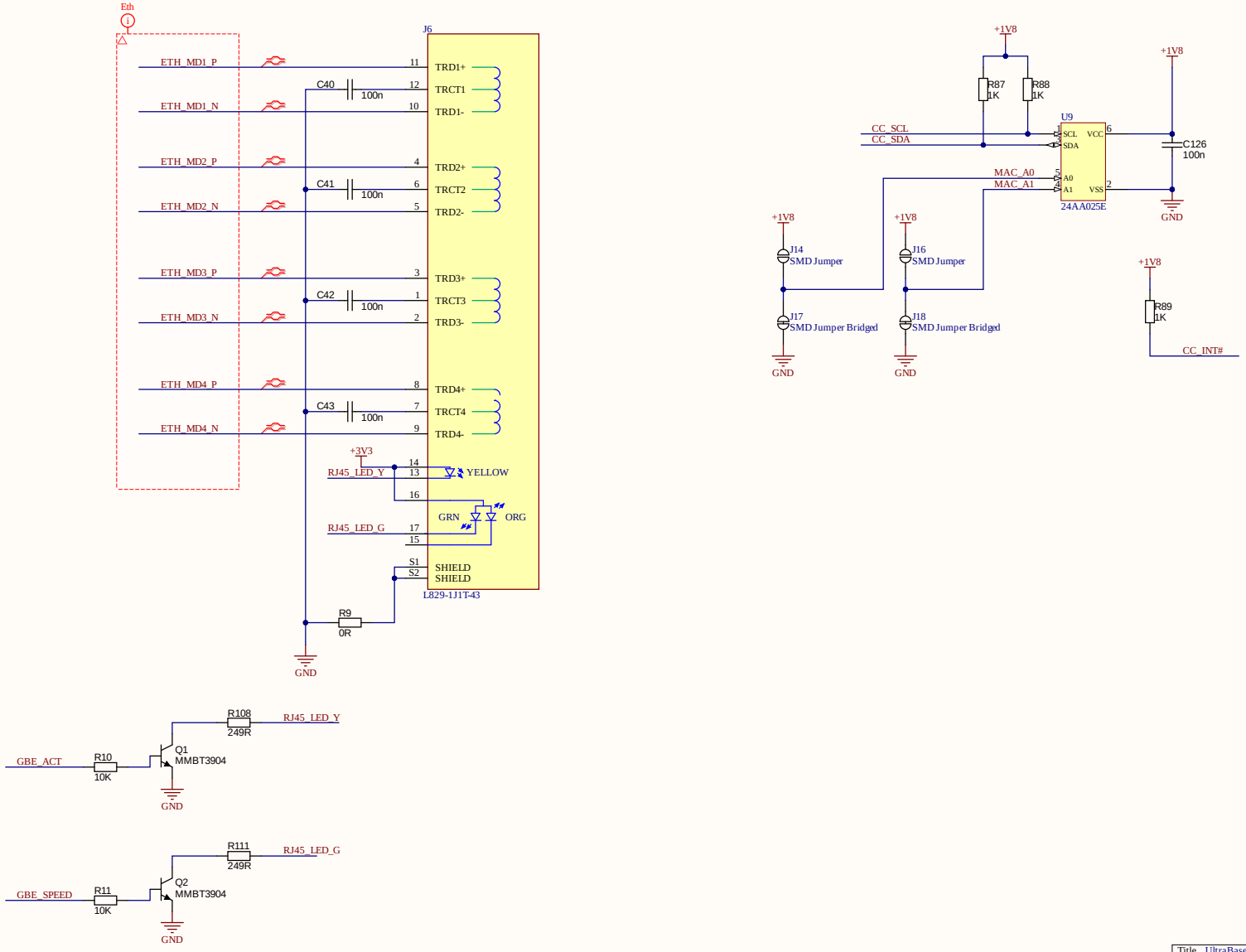

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|---|---|---|
| Title | UltraBase V2.1 5424<br>PS Connector<br>www.CircuitValley.com | |
| Size<br>A3 | Number | Revision |
| Date: 6-21-2025 | | Sheet of |
| File: C:\Users\..\SOM_JX_MIO.SchDoc | | Drawn By: |



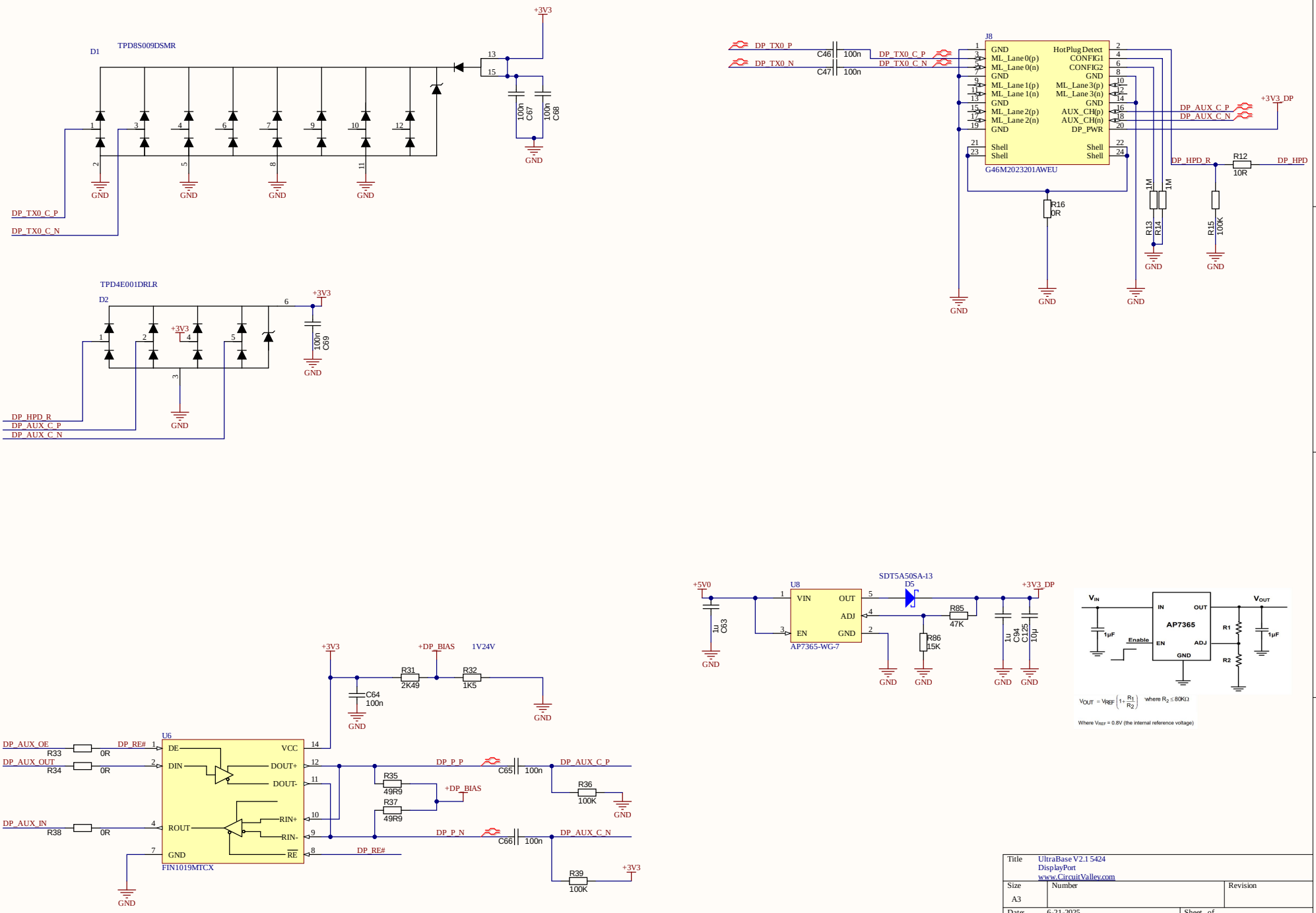




| | | |
|---|---|---|
| Title | UltraBase V2.1 5424<br>MIO SD<br>www.CircuitValley.com | |
| Size<br>A3 | Number | Revision |
| Date: 6-21-2025 | | Sheet of |
| File: C:\Users\..\MIO_SD.SchDoc | | Drawn By: |

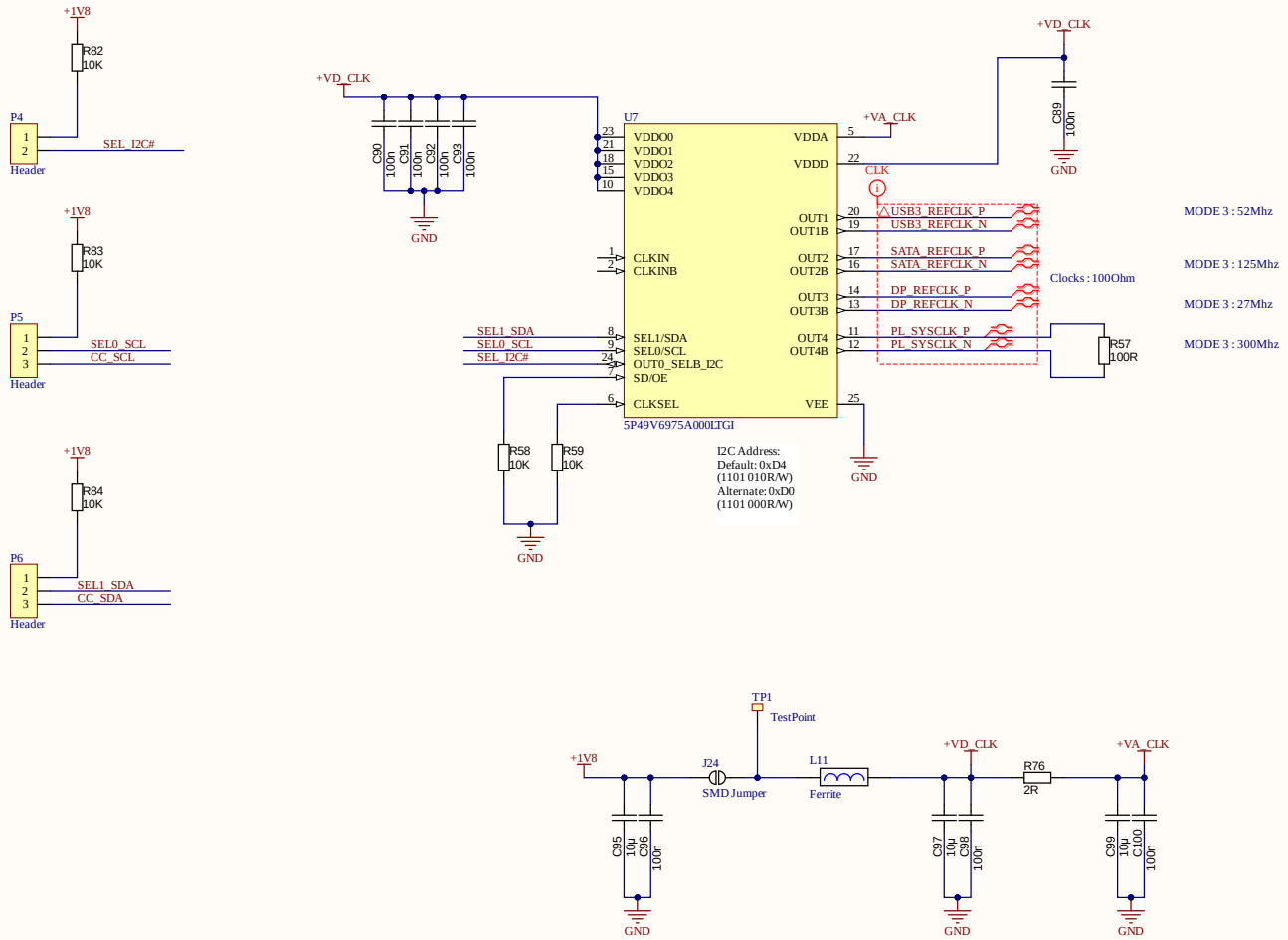

U1
MIC29302WU
MIC29152/302/502
U5A
I2C Interface w variable Ref.
IRPS5401M
Change Package
User's Note:
- I2C Address Offset +2
(Set by R on pin 55 = 1.78K)
U5C
SWITCHER A – 2A
SWITCHER B – 2A
U5B
Sequencing & Ramp Control
Tracking/Non-Tracking LDO
Timing & Phase
Registers and MTP Memory
User's Note:
- MTP Bank# set by R on PIn 54
- Switching Frequency: 1MHz
U5D
SWITCHER C – 4A
SWITCHER D – 4A
Vout = 0.5 ( 1+ (RFB/RGND))
Title UltraBase V2.1 5424
Power Regulator
www.CircuitValley.com
Size A3
Number
Revision
Date: 6-21-2025
File: C:\Users\..\POWER.SchDoc
Sheet of
Drawn By:



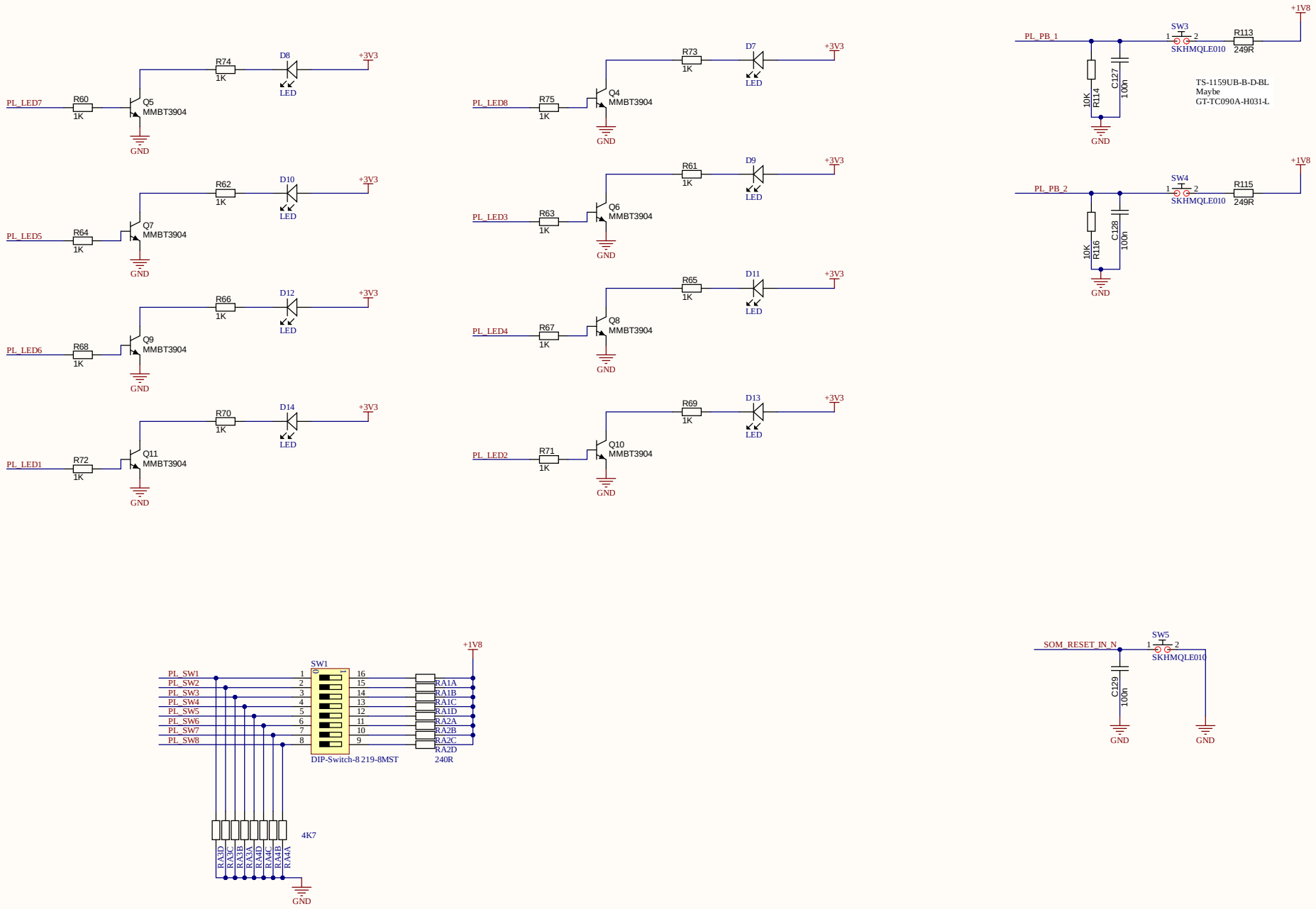

| | | |
|---|---|---|
| Title | UltraBase V2.1 5424 Ethernet<br>www.CircuitValley.com | |
| Size: A3 | Number | Revision |
| Date: 6-21-2025 | | Sheet of |
| File: C:\Users\..\Ethernet.SchDoc | | Drawn By: |



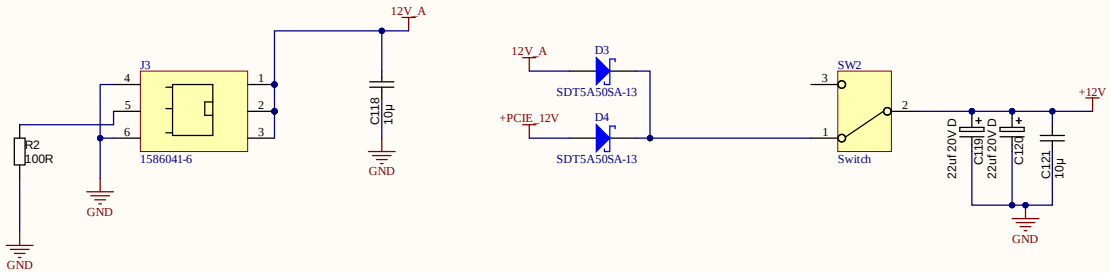

| | | |
|---|---|---|
| Title | UltraBase V2.1 5424<br>DisplayPort<br>www.CircuitValley.com | |
| Size<br>A3 | Number | Revision |
| Date: | 6-21-2025 | Sheet of |
| File: | C:\Users\..\DisplayPort.SchDoc | Drawn By: |




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|---|---|---|
| Title | UltraBase V2.1 5424<br>Clock Generator<br>www.CircuitValley.com | |
| Size<br>A3 | Number | Revision |
| Date: | 6-21-2025 | Sheet of |
| File: | C:\Users\..\Clock.SchDoc | Drawn By: |

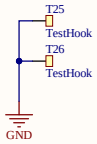


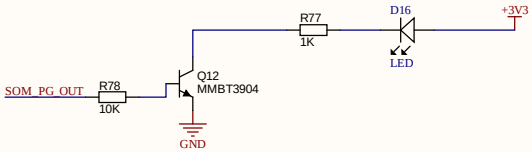

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|---|---|---|
| Title | UltraBase V2.1 5424<br>Switches and LEDs<br>www.CircuitValley.com | |
| Size<br>A3 | Number | Revision |
| Date: 6-21-2025 | | Sheet of |
| File: C:\Users\..\Switches_Leds.SchDoc | | Drawn By: |

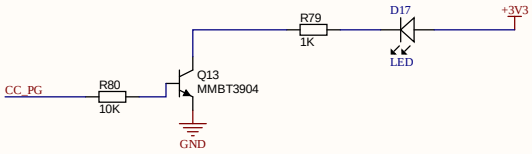


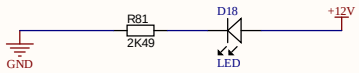


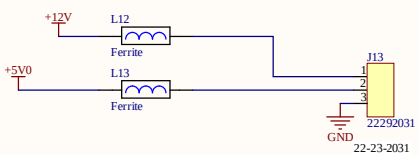


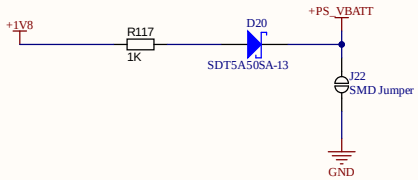


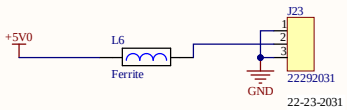



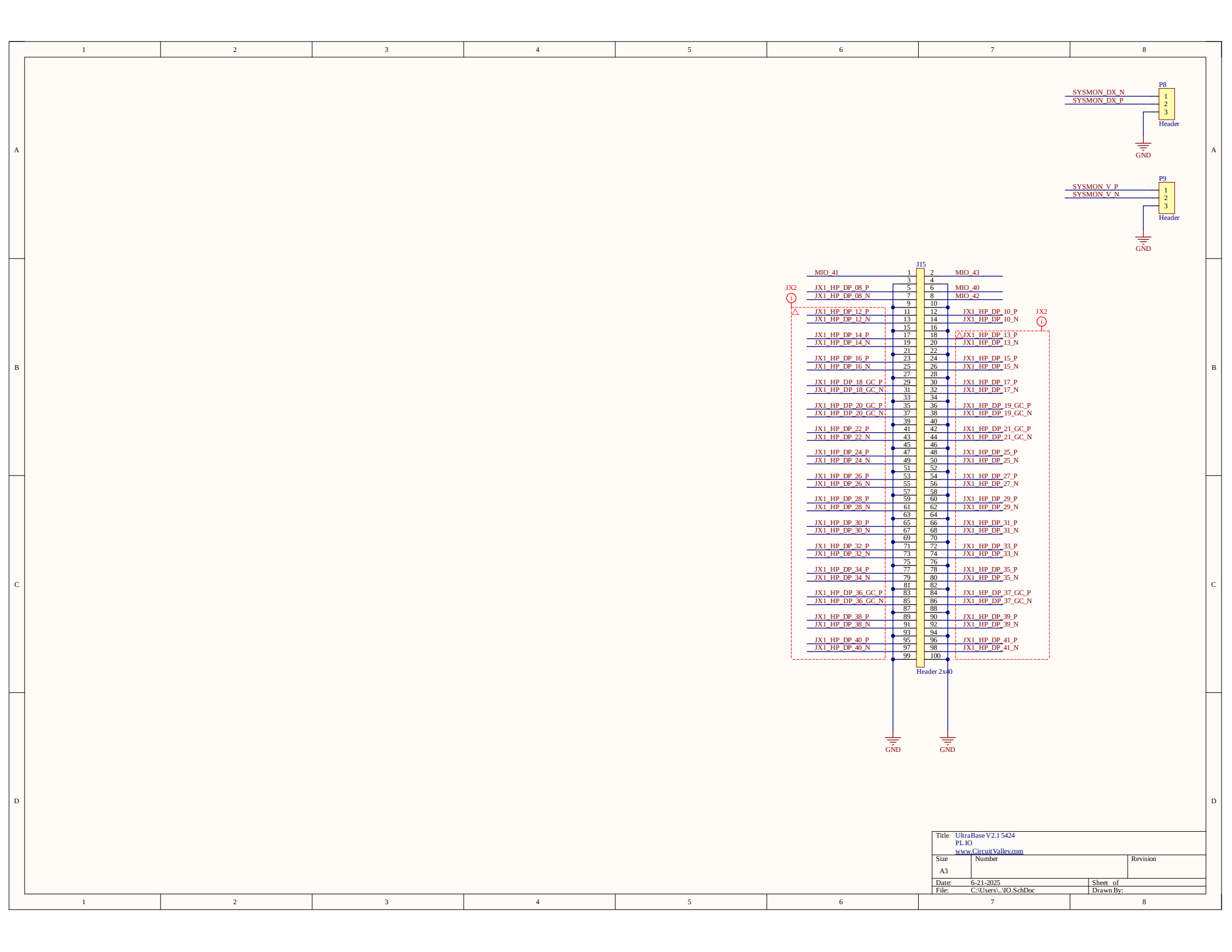



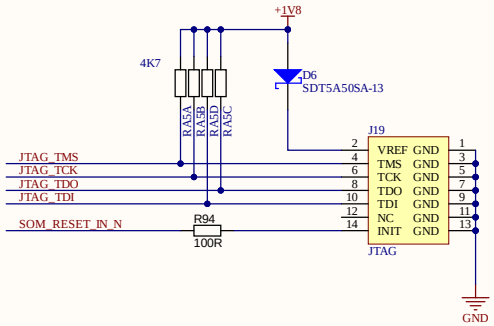



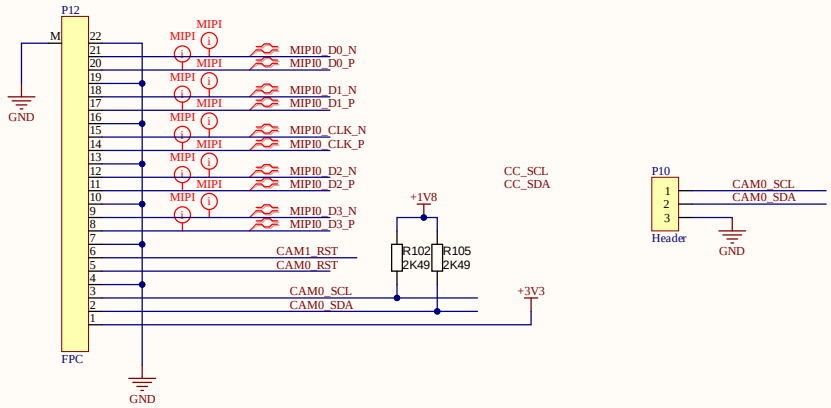




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|---|---|---|
| Title | UltraBase V2.1 5424<br>Power Input<br>www.CircuitValley.com | |
| Size<br>A3 | Number | Revision |
| Date: | 6-21-2025 | Sheet of |
| File: | C:\Users\..\PWR_INP.SchDoc | Drawn By: |


P8
SYSMON_DX_N
SYSMON_DX_P
1
2
3
Header
GND
P9
SYSMON_V_P
SYSMON_V_N
1
2
3
Header
GND
J15
JX2
JX2
MIO_41
MIO_43
JX1_HP_DP_08_P
JX1_HP_DP_08_N
MIO_40
MIO_42
JX1_HP_DP_12_P
JX1_HP_DP_12_N
JX1_HP_DP_10_P
JX1_HP_DP_10_N
JX1_HP_DP_14_P
JX1_HP_DP_14_N
JX1_HP_DP_13_P
JX1_HP_DP_13_N
JX1_HP_DP_16_P
JX1_HP_DP_16_N
JX1_HP_DP_15_P
JX1_HP_DP_15_N
JX1_HP_DP_18_GC_P
JX1_HP_DP_18_GC_N
JX1_HP_DP_17_P
JX1_HP_DP_17_N
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JX1_HP_DP_20_GC_N
JX1_HP_DP_19_GC_P
JX1_HP_DP_19_GC_N
JX1_HP_DP_22_P
JX1_HP_DP_22_N
JX1_HP_DP_21_GC_P
JX1_HP_DP_21_GC_N
JX1_HP_DP_24_P
JX1_HP_DP_24_N
JX1_HP_DP_25_P
JX1_HP_DP_25_N
JX1_HP_DP_26_P
JX1_HP_DP_26_N
JX1_HP_DP_27_P
JX1_HP_DP_27_N
JX1_HP_DP_28_P
JX1_HP_DP_28_N
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JX1_HP_DP_31_P
JX1_HP_DP_31_N
JX1_HP_DP_32_P
JX1_HP_DP_32_N
JX1_HP_DP_33_P
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JX1_HP_DP_34_P
JX1_HP_DP_34_N
JX1_HP_DP_35_P
JX1_HP_DP_35_N
JX1_HP_DP_36_GC_P
JX1_HP_DP_36_GC_N
JX1_HP_DP_37_GC_P
JX1_HP_DP_37_GC_N
JX1_HP_DP_38_P
JX1_HP_DP_38_N
JX1_HP_DP_39_P
JX1_HP_DP_39_N
JX1_HP_DP_40_P
JX1_HP_DP_40_N
JX1_HP_DP_41_P
JX1_HP_DP_41_N
Header 2x40
GND
GND
Title UltraBase V2.1 5424
PL IO
www.CircuitValley.com
Size A3
Number
Revision
Date: 6-21-2025
Sheet of
File: C:\Users\..\IO.SchDoc
Drawn By:




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|---|---|---|
| Title | UltraBase V2.1 5424<br>JTAG<br>www.CircuitValley.com | |
| Size<br>A3 | Number | Revision |
| Date: 6-21-2025 | | Sheet of |
| File: C:\Users\..\JTAG.SchDoc | | Drawn By: |




| | | |
|---|---|---|
| Title | UltraBase V2.1 5424<br>MIPI<br>www.CircuitValley.com | |
| Size<br>A3 | Number | Revision |
| Date: 6-21-2025 | | Sheet of |
| File: C:\Users\..\MIPI.SchDoc | | Drawn By: |